

High fidelity modeling of pumped storage units for optimal operation of a multi-energy co-generation system

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ABSTRACT

Pumped storage, as the storage technology with the largest installed capacity and mature technology, plays a key regulation role in the multi-energy co-generation system. The core of regulation accuracy is the high-fidelity modeling of pumped storage. This paper improves the insufficiency of the conventional pumped storage model (CPSM) in exaggerating the flexibility of pumped storage units by setting up non-operable operating condition regions, and incorporates the condition switching loss into the model consideration, on the basis of which a high fidelity pumped storage model (HFPSM) is established. Secondly, a multi-objective function of the operation model is set from the consideration of economy, environmental protection and stability, and a engineering application level optimal operation model of wind power-photovoltaic-thermal power-pumped storage multi-energy co-generation system based on HFPSM is proposed. Thirdly, two research indicators, the thermal power volatility indicator and the pumped storage unit utilization rate, are proposed. Fourthly, combined with examples, the effects of CPSM and HFPSM on the optimized operation of the system are investigated. It is found that adding four 50MW pumped storage units to the system can reduce the total operating cost by 10.34% and the environmental cost by 83.61%. The optimization of the high fidelity model is better under the premise of ensuring safe, stable and efficient operation of the system. Optimizing the number of unit startups and shutdowns for the high fidelity model to 25.8% of the conventional model. Finally, by exploring the optimal operation of a multi-energy co-generation system with different pumped storage installed capacities, it is found that the proposed high fidelity model makes the optimization results more in line with the actual operation. The flexibility of the pumped storage units no longer increases with the increase of installed capacity, which provides a reference for the planning of the storage capacity in the multi-energy co-generation system.

1. Introduction

With the depletion of fossil fuels and growing concern about the impact on the environment, accelerating the development of renewable energy has become an inevitable choice for energy transformation (Wang et al., 2023a; Mahfoud et al., 2023). Renewable energy generation has the characteristics of stochasticity, intermittency and difficulty in scheduling, large-scale use of renewable energy generation brings new challenge to the power system (Daneshvar et al., 2020).

The inclusion of storage in the power system can provide greater flexibility for the system, stabilize the volatility of renewable energy, promote the consumption of renewable energy, and to some extent eliminate the negative impacts of the large-scale use of renewable energy. Pumping storage as the largest installed capacity, the most mature technology of energy storage technology, is the focus of the

current research (Hunt et al., 2020). Investigating the optimal operation of multi-energy co-generation system adding pumped storage can take full advantage of the flexibility of pumped storage, consume more renewable energy, and reduce carbon emission. However, the research premise is high fidelity modeling of the power system (Lu et al., 2022).

Research on the optimal operation of multi-energy co-generation systems for adding pumped storage is a hot issue in power systems which has been given a lot of attention in recent years. In these studies, some scholars devote themselves to the study of model solving algorithms. Xu et al. (2020), Patwal et al. (2018) and Zhang et al. (2023) used the improved particle swarm algorithm and the imperialistic competitive algorithm to solve the hybrid energy system model, and compared them with other algorithms to show the superiority of the proposed algorithms. Basu and Das (2023) and Jena et al. (2022)

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Nomenclature

T	dispatch cycle (24 h)
C_c	System power generation cost
$p_{hp,t}$	Pumping price for t-th period
$P_{load,t}$	load in t-th period
Δt	length of a period of time (0.25 h)
$Q_{c,q}$	actual carbon emissions
γ	hydroelectricity conversion constant
I_v	Indicators of TP output volatility
C_c^{hp}	Pumping power consumption cost
δ	coefficient of unit power emission quota
$U_{t,n}^{hh}, U_{t,n}^{hp}$	binary status indicator of power generation and pumping of PS unit
$P_{t,n}^{hh}, P_{t,n}^{hp}$	Generating power and pumping power of the n-th PS unit in t-th period
$T_{run/offset-min}^{hh}, T_{run/offset-min}^{hp}$	lower limit of startup/shutdown time for continuous power generation and pumping of PS unit
N_h, N_f, N_w, N_p	number of PS, TP, WP and PV units
U_{h-h-t}, U_{h-p-t}	binary status indicator of power generation and pumping of power station
$P_{G,n}^{hh}, P_{G,n}^{hp}$	rated power of PS unit under power generation and pumping condition
$\bar{P}_{t,n}^{hh}, \bar{P}_{t,n}^{hp}$	Generating and pumping power without considering the condition conversion loss
c_p, c_w, c_{hh}	Unit power generation cost coefficient for PV, WP, and PS unit
$T_{run/offset}^{hh}, T_{run/offset}^{hp}$	continuous power generation and pumping operation / shutdown time of PS unit
$P_{min,n}^f, P_{max,n}^f$	Upper and lower bounds of the output of the n-th TP unit
$P_{down,n}^f, P_{up,n}^f$	Ramp up/down limits value of the n-th TP unit
$P_{min,n}^p, P_{max,n}^p$	Upper and lower bounds of the output of the n-th PV unit
$E_{t,n,loss}^{hh}, E_{t,n,loss}^{hp}$	Generation and pumping energy loss of PS unit during condition conversion
$V_{min}^{up}, V_{min}^{down}$	minimum storage capacity of upper reservoir / lower reservoir
ΔV	Water volume changes during the scheduling period
μ_{hh}, μ_{hp}	generation efficiency and pumping efficiency of PS unit
T_{f-run}, T_{f-off}	continuous power generation operation / shutdown time of TP unit
I_{use}^h	utilization rate of PS units
ΔQ_c	Interval length for ladder carbon trading
H	net head

C_m	System equipment maintenance cost
C_{c-out}	cost of carbon emission
C_e	environmental benefits
p_{c-base}	carbon trading base price
k	price coefficient
c_{e-p}, c_{e-w}	environmental benefit coefficient
y	random number
C_{loss}^p, C_{loss}^w	Cost of solar and wind curtailment
η	actual carbon emission coefficient
$U_{t,n}^f, U_{t,n}^p, U_{t,n}^w$	binary status indicator of TP, PV and WP unit
$C_{run-off}$	startup / shutdown cost of thermal power
$P_{min,n}^{hh}, P_{max,n}^{hh}$	upper and lower bounds of the power generation condition of the n-th PS unit
$P_{t,n}^{p-pre}, P_{t,n}^{w-pre}$	The predicted output of PV and WP unit
$P_{f,t,n}^p, P_{t,n}^p, P_{t,n}^w$	Generating power of the n-th TP, PV and WP unit in t-th period
c_{f2}, c_{f1}, c_f	coefficient of unit power generation cost for TP unit
$P_{down,n}^{hh}, P_{up,n}^{hh}$	Ramp up/down limits value of the power generation of the PS unit
$c_{m-f}, c_{m-p}, c_{m-w}, c_{m-h}$	cost coefficient for maintenance of TP, PV, WP and PS unit equipment
$T_{f-run,n}^{min}, T_{f-off,n}^{min}$	lower limit of continuous power generation operation / shutdown time of TP unit
$C_{run-off}$	TP unit startup/shutdown cost coefficient
$P_{min,n}^w, P_{max,n}^w$	Upper and lower bounds of the output of the n-th WP unit
c_{p-pen}, c_{w-pen}	solar and wind curtailment cost coefficient
$Q_{c,r}$	quota of carbon emissions allocated by the government
$V_{max}^{up}, V_{max}^{down}$	maximum storage capacity of upper reservoir / lower reservoir
$\Delta \bar{V}$	upper limit of water volume change of power station
λ	Cost coefficient for output volatility of TP units
$N_{run-off}$	number of TP unit startups/shutdowns
I_{w-c}	wind curtailment rate

improved the genetic algorithm for solving the system model. Patwal and Narang (2020) and Gao et al. (2022) presented a new solution method for hydrothermal system and verified the superiority of the proposed method. Diab et al. (2020) used Whale Optimization Algorithm to size the hybrid system. Feng et al. (2021) used CPLEX called by Matlab to simulate and solve the scheduling model. Some scholars have

attempted to incorporate deep learning to improve research accuracy and efficiency. Yang et al. (2023) combined neural networks and long and short-term memory autoencoders to optimize the distributed proximal policy of the system. Wang et al. (2023c) proposed a wind-pumped storage-hydrogen storage combined operation system based on deep learning and intelligent optimization. For increased system flexibility, some scholars have combined pumped storage with chemical energy storage. Abdelshafy et al. (2020) introduced a grid-connected double storage system consisting of pumped storage and battery. Li et al. (2018) designed a practical framework to optimize the operation of a hybrid system consisting of wind, pumped storage and battery and demonstrated the advantages of the proposed coordination method through three cases. Yadav et al. (2023) applied the grey wolf algorithm to optimize a double storage isolated microgrid with battery and pumped hydro which significantly reduced the greenhouse gas emissions of the system. Javed et al. (2020) proposed a hybrid pumped and battery storage (HPBS) system to make the off-grid RE system more reliable and sustainable. For optimal scheduling research, it is also a good entry point to start with electricity price. Shyam and Kanakasabapathy (2022), Kumar et al. (2022) and Nasab et al. (2022)

investigated hybrid energy systems in terms of electricity markets and tariffs

In the above researches, energy storage is an indispensable component, and pumped storage as the most common one is the focus of current research. Basu (2020), Sun et al. (2019) and Ma et al. (2022) used pumped storage in a hybrid energy system to improve the economics of the system. Li et al. (2023b), Djelailia et al. (2019) and Ma et al. (2020) reduced the carbon emissions of a multi-energy co-generation system by optimizing a multi-energy co-generation system with pumped storage. Yuan et al. (2022) and Özyön (2020) investigated the short-term optimal operation of the system by processing the constraints and objective functions in the hybrid energy system model through deterministic transformations, linearization techniques and penalty functions. Cheng et al. (2018) developed a model based on Mixed-integer linear programming (MILP) to determine the optimal hourly scheduling of pumped-storage hydropower plants serving multiple provincial power grids. Li et al. (2023a) proposed an operation mode of priority regulation of pumped storage (PS) for hybrid energy system. Wang et al. (2023b) proposed a novel assessment model to assess the regulation intensity of the pumped storage unit for variable renewable energy consumption. Bafrani et al. (2022) proposed a stochastic optimization model to optimally operate thermal power plants together with hydro pumped storage units in day ahead market of electrical reserve and energy. Salimi et al. (2019) confirmed the effectiveness of the adjustable speed pumped-storage hydropower plants device by adding various numerical studies on the penetration factor of wind power plants. Liang et al. (2020) investigated the optimal operation of an island microgrid containing wind, photovoltaic and diesel generators using seawater pumped storage power station as energy storage equipment. Despite the fact that many fruitful works have been accomplished by scholars for the optimal operation of multi-energy co-generation system, some minor problems have been identified. There is a slight lack of accurate research for operational modeling, especially for pumped storage, the regulator in multi-energy co-generation system. In the above references, the modeling for pumped storage unit is relatively simplified, the model does not take into account the actual operating characteristics of the unit and the losses of the unit's switching of operating conditions. Meanwhile, most of the mentioned studies are hour-level optimization operation models, there is a lack of engineering application level optimization operation models for smaller time scales.

In the background of energy revolution in recent years, the development of renewable energy is becoming more and more rapid. China is one of the representatives, both the annual renewable energy installed capacity and the total installed capacity of renewable energy ranked first in the world. The energy regulation model proposed in this paper can significantly improve the consumption of renewable energy, meanwhile, accurately quantify the regulation capacity of pumped storage. In some areas of western China, the installed capacity of renewable energy is more than 70%, and it is urgent to have an accurate and detailed energy regulation model. Secondly, pumped storage is an important component as a regulator of multi-energy co-generation, and the existing models are not detailed enough in describing the pumped storage regulation model. Therefore, it is of great engineering interest to study the optimal operation of multi-energy co-generation system based on high fidelity modeling of pumped storage.

To address the abovementioned issues, in this paper, an engineering application level optimization operation model of wind power (WP)-photovoltaic (PV)-thermal power (TP)-pumped storage (PS) multi-energy co-generation system based on pumped storage high fidelity modeling is proposed with the contributions shown below.

- This paper accurately describes the operating range of pumped storage units by setting the non-operable operating condition regions in combination with the actual operating characteristics of pumped storage units.

- This paper considers the unit combinations while accurately quantifying the time loss and power loss when the units switch operating conditions. Based on this, a high fidelity energy regulation model for pumped storage units is proposed.
- From the consideration of economy, environmental protection and stability, a multi-objective function is set for the operation model.
- The example results show that the proposed high fidelity energy regulation model reduces the number of pumped storage units operating condition switching by 74.2% compared with the conventional model under the premise of ensuring the safe and stable operation of the system.

The remainder of the study is structured as follows. Section 2 introduces a engineering application level optimal operation model of multi-energy co-generation system based on pumped storage high fidelity modeling; Section 3 describes the basics of the simulation example; Section 4 introduces the results of the simulation; Section 5 presents the conclusions and future work.

2. Optimal operation model

The multi-objective optimal operation of a multi-energy co-generation system is a substantial problem, by utilizing the coupling mechanism of each energy source in space and time, to realize multi-energy complementarity and achieve a high proportion of renewable energy consumption while taking into account the system stability, safety and other operational objectives. In this paper, the multi-energy co-generation system consists of power grid, customer load, WP, PV, TP and PS, as shown in Fig. 1. WP, PV and TP are responsible for power production and supplying energy to the customers through the power grid. Pumped storage acts as a regulator of the power generation system, regulating the energy of the grid by pumping and discharging water for power generation to achieve the effect of peak-cutting and valley-filling.

2.1. High fidelity modeling of PS

The research uses pumped storage units as risk management energy storage devices for multi-energy co-generation system, modeled as follows.

The operating range of the unit is a critical factor in the optimized operation model. In the previous optimized operation model, CPSM sets the operating condition can be any point between the rated generating power and the rated pumping power, which is inconsistent with the actual operating characteristics of the pumped storage unit. When the pumped storage unit generates electricity at small output, the unit is extremely unstable under the action of hydraulic and mechanical factors, operation in this state for a long time will cause damage to the unit. In pumping condition, conventional pumped storage units can only pump at rated condition and cannot change the pumping power. Variable-speed pumped storage units also only change power within a certain range. Since most pumped storage units are conventional pumped storage units, the research object of this paper is conventional pumped storage units. This paper proposes to set the non-operable operating condition region of pumped storage units in the model to improve the previous CPSM. For expressing the state of pumped storage power station and each unit, binary variables are introduced as state indicators, when the value of the power generation condition state indicator is 1, it indicates that the pumped storage unit is releasing water for power generation, when its value is 0, the pumped storage unit is not generating power. similarly, the pumping condition indicator and the pumped storage power station state indicator are set. The operating range is expressed as

$$U_{t,n}^{hh} P_{\min,n}^{hh} \leq P_{t,n}^{hh} \leq U_{t,n}^{hh} P_{\max,n}^{hh} \quad (1)$$

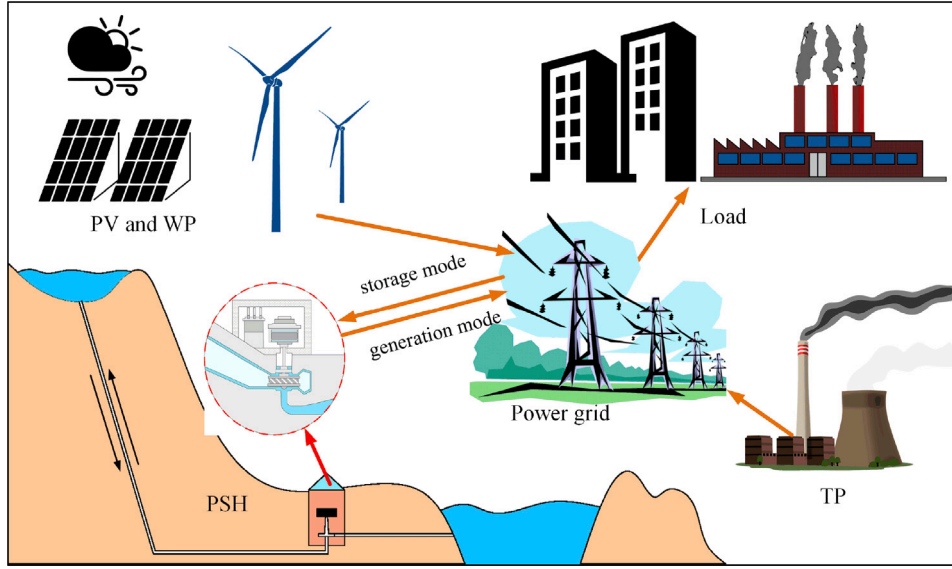


Fig. 1. Multi-energy co-generation system.

$$U_{t,n}^{hp} P_{G,n}^{hp} \leq P_{t,n}^{hp} \leq U_{t,n}^{hp} P_{G,n}^{hp} \quad (2)$$

where the lower limit of the operating range of the power generation condition of the pumped storage unit $P_{\min,n}^{hh}$, is expressed as $P_{\min,n}^{hh} = a P_{\max,n}^{hh}$, a is the proportion coefficient of the unit generation non-operable operating condition relative to the rated generating power. The value of $P_{\max,n}^{hh}$ is generally taken as the unit's rated generating power $P_{G,n}^{hh}$. After setting the non-operable operating condition region in the model, the range of pumped storage power station generating condition is represented as a superposition of intervals of ranges, and the operating range of pumping condition is represented as a stepped set of points. Conventional pumped storage power stations, in actual operation, generally do not appear a power station at the same time the phenomenon of pumping and power generation, much less a unit at the same time pumping and power generation (Feng et al., 2021).

$$U_{t,n}^{hh} + U_{t,n}^{hp} \leq 1 \quad (3)$$

$$\sum_{n=1}^{N_h} U_{t,n}^{hh} \leq N_h U_{h-h-t} \quad (4)$$

$$\sum_{n=1}^{N_h} U_{t,n}^{hp} \leq N_h U_{h-p-t} \quad (5)$$

$$U_{h-h-t} + U_{h-p-t} \leq 1 \quad (6)$$

Research considers optimized operation at the engineering application level (15-minute level), and in this small time scale optimized operation investigation, the effect of pumped storage unit condition switching on the optimized operation cannot be ignored. Taking the pumped storage unit switching from power generation to pumping condition as an example, the pumped storage unit firstly has to unload the load, disconnect from the grid, reduce the unit speed, close the guide vane, shut down the unit, then the unit connects to the grid again, turn on the pumping condition and a series of other operations, during which the output of the unit is constantly changing. This process has some time loss and power loss for grid scheduling. For this problem, the research uses the power loss to quantify the time loss and power loss at the time of condition switching in a unified way. The power loss in this paper refers to the time-power curve of Tianhuangping Pumped

Storage Power Station during the switching of operating conditions. It is represented in the model as follow

$$\begin{cases} P_{t,n}^{hh} = \frac{\Delta t \bar{P}_{t,n}^{hh} - E_{t,n,loss}^{hh}}{\Delta t} & U_{t,n}^{hh} = 1 \text{ and } U_{t+1,n}^{hh} = 0 \\ P_{t+1,n}^{hh} = \frac{\Delta t \bar{P}_{t+1,n}^{hh} - E_{t+1,n,loss}^{hh}}{\Delta t} & U_{t,n}^{hh} = 0 \text{ and } U_{t+1,n}^{hh} = 1 \\ P_{t,n}^{hp} = \frac{\Delta t \bar{P}_{t,n}^{hp} - E_{t,n,loss}^{hp}}{\Delta t} & U_{t,n}^{hp} = 1 \text{ and } U_{t+1,n}^{hp} = 0 \\ P_{t+1,n}^{hp} = \frac{\Delta t \bar{P}_{t+1,n}^{hp} - E_{t+1,n,loss}^{hp}}{\Delta t} & U_{t,n}^{hp} = 0 \text{ and } U_{t+1,n}^{hp} = 1 \end{cases} \quad (7)$$

where $E_{t,n,loss}^{hh}$ and $E_{t,n,loss}^{hp}$ are the energy loss of the pumped storage unit for the switching of operating conditions in the power generation and pumping conditions, respectively, denoted as $E_{t,n,loss}^{hh} = s_{hh} \bar{P}_{t,n}^{hh}$ and $E_{t,n,loss}^{hp} = s_{hp} \bar{P}_{t,n}^{hp}$, where s_{hh} and s_{hp} are the loss coefficients, which are obtained by referring to the time-power curve of Tianhuangping Pumped Storage Power Station during the switching of operating conditions.

In the operation of pumped storage power stations, some reservoirs also have to take into account issues such as fishery farming, ecology and agricultural irrigation, and there are certain limitations in the capacity of the upper and lower reservoirs, which are linked together through hydraulics and electricity (Cheng et al., 2018).

$$\begin{cases} V_{\min}^{up} \leq V_t^{up} \leq V_{\max}^{up} \\ V_{\min}^{down} \leq V_t^{down} \leq V_{\max}^{down} \end{cases} \quad (8)$$

where V_t^{up} and V_t^{down} are the real-time capacities of the upper and lower reservoirs, respectively, which can be obtained by using equation $V_t^{up} = V_{t-1}^{up} - \sum_{n=1}^{N_h} \int \frac{\mu_{hh} P_{t,n}^{hh}}{\gamma H} + \frac{P_{t,n}^{hp}}{\gamma H \mu_{hp}} dt$ and equation $V_t^{down} = V_{t-1}^{down} + \sum_{n=1}^{N_h} \int \frac{\mu_{hh} P_{t,n}^{hh}}{\gamma H} + \frac{P_{t,n}^{hp}}{\gamma H \mu_{hp}} dt$, the maximum and minimum capacities of the upper and lower reservoirs are determined in accordance with arrangements made by the reservoir scheduling department. It is also important to note that PS, as a risk management energy storage device in a multi-energy co-generation system, should achieve a basic balance between the amount of water used for power generation and pumping over an operating cycle.

$$\Delta V \leq \Delta \bar{V} \quad (9)$$

$$\Delta V = \sum_{t=1}^T \sum_{n=1}^{N_h} \frac{\mu_{hh} P_{t,n}^{hh}}{\gamma H} + \frac{P_{t,n}^{hp}}{\gamma H \mu_{hp}} \quad (10)$$

Research considers optimized operation at the engineering application level with small time scales, so certain requirements should be made for the ramp up/down of the pumped storage units. Meanwhile, because the optimized operation is a continuous operation, the end time period of the previous operating cycle connects to the start time period of the next operating cycle, and considering the similarity between one cycle and another, the ramp up/down limit and the start/end power limit are set as follows (Bafrani et al., 2022)

$$P_{down,n}^{hh} \leq P_{t,n}^{hh} - P_{t-1,n}^{hh} \leq P_{up,n}^{hh} \quad (11)$$

$$P_{down,n}^{hh} \leq P_{1,n}^{hh} - P_{T,n}^{hh} \leq P_{up,n}^{hh} \quad (12)$$

Under the premise of ensuring the safe and stable operation of the system, the optimized operation should try to ensure the stable operation of the unit, so it is significant to set the minimum operation time of the unit.

$$\begin{cases} T_{run/off}^{hh} \geq T_{run/off-min}^{hh} \\ T_{run/off}^{hp} \geq T_{run/off-min}^{hp} \end{cases} \quad (13)$$

CPSM only included the unit's operating range constraint, unit state constraint, water constraint, and cycle water balance constraint. Including Eqs (1)–(6) and (8)–(10). And for the value of unit operating range constraint $P_{min,n}^{hh}$ is 0, Eq. (2) becomes $U_{t,n}^{hp} P_{G,n}^{hp} \leq P_{t,n}^{hp} \leq 0$. In this paper, the high fidelity model establishes the non-operable operating condition regions, which improves the operating range of the pumped storage unit, and takes the power loss and time loss during the switching of the operating conditions into consideration; Under the premise of ensuring safe, stable and economic operation of the power grid and considering the stable operation of pumped storage units, the minimum startup and shutdown time are proposed. These improvements relative to CPSM ensure the accuracy and stability of the optimized operation of pumped storage units.

2.2. Objective function

In the new power system context, the grid should not only consider economics and security, but also factors such as carbon emissions, renewable energy consumption and environmental benefits. Therefore, this paper proposes objective functions O_1 , O_2 and O_3 from three aspects of economy, environment and system stability. The total objective function O is shown in Eq. (14).

$$\min O = O_1 + O_2 + O_3 \quad (14)$$

2.2.1. Economic objective function

The economic object O_1 is the total operation cost in the whole operation cycle, mainly considering the generation cost of each power unit C_c , equipment operation and maintenance cost C_m , TP startup/shutdown cost $C_{run-off}$, pumped storage power station pumped power consumption cost C_c^{hp} , solar curtailment cost C_{loss}^p and wind curtailment cost C_{loss}^w , are expressed as

$$O_1 = C_c + C_m + C_{run-off} + C_c^{hp} + C_{loss}^p + C_{loss}^w \quad (15)$$

where the generation cost of each power unit C_c and equipment operation and maintenance cost C_m are expressed as (Lu et al., 2022)

$$\begin{aligned} C_c = & \sum_{t=1}^T \sum_{n=1}^{N_f} (c_{f2} P_{f,t,n}^2 + c_{f1} P_{f,t,n} + c_f) + \sum_{t=1}^T \sum_{n=1}^{N_p} (c_p P_{t,n}^p) \\ & + \sum_{t=1}^T \sum_{n=1}^{N_w} (c_w P_{t,n}^w) + \sum_{t=1}^T \sum_{n=1}^{N_h} (c_{hh} P_{t,n}^{hh}) \end{aligned} \quad (16)$$

$$C_m = \sum_{t=1}^T \sum_{n=1}^{N_f} (c_{m-f} P_{f,t,n}) + \sum_{t=1}^T \sum_{n=1}^{N_p} (c_{m-p} P_{t,n}^p)$$

$$+ \sum_{t=1}^T \sum_{n=1}^{N_w} (c_{m-w} P_{t,n}^w) + \sum_{t=1}^T \sum_{n=1}^{N_h} [c_{m-h} (P_{t,n}^{hh} - P_{t,n}^{hp})] \quad (17)$$

In actual operation, TP startup and shutdown costs are expensive, and the study takes them into account, denoted as

$$C_{run-off} = \sum_{n=1}^{N_h} (c_{run-off} N_{run-off}) \quad (18)$$

Peak and valley tariff is introduced in the research to calculate the pumped storage power station pumped power consumption cost, C_c^{hp} as shown in Eq. (19).

$$C_c^{hp} = \sum_{t=1}^T \sum_{n=1}^{N_h} (P_{hp,t} P_{t,n}^{hp}) \quad (19)$$

For improving the proportion of renewable energy consumption, the solar curtailment cost C_{loss}^p and the wind curtailment cost C_{loss}^w are set as shown in Eq. (20) and (21) (Kumar et al., 2022).

$$C_{loss}^p = \sum_{t=1}^T \sum_{n=1}^{N_p} (c_{p-pen} (P_{t,n}^{p-pre} - P_{t,n}^p)) \quad (20)$$

$$C_{loss}^w = \sum_{t=1}^T \sum_{n=1}^{N_w} (c_{w-pen} (P_{t,n}^{w-pre} - P_{t,n}^w)) \quad (21)$$

2.2.2. Environmental objective function

Environmental objective function O_2 considers the optimal operation of the multi-energy co-generation system from the perspective of environmental protection. Considering the environmental benefits of renewable energy sources in addition to quantifying the carbon emissions in the power generation process.

$$O_2 = C_{e-out} + C_e \quad (22)$$

The study introduces factors such as stepped carbon trading and government-allocated carbon exemptions to quantify the cost of carbon emission from TP in the power generation process, which is expressed as Eq. (23) (see Box I). where the government's allocation of CO_2 emission allowances $Q_{c,r}$ and the actual carbon emissions $Q_{c,q}$ are calculated by equation $Q_{c,r} = \sum_{t=1}^T \sum_{n=1}^{N_f} (\delta P_{f,t,n})$ and equation $Q_{c,q} = \sum_{t=1}^T \sum_{n=1}^{N_f} (\eta P_{f,t,n})$. The environmental benefits of renewable energy are expressed as

$$C_e = \sum_{t=1}^T \sum_{n=1}^{N_p} (c_{e-p} P_{t,n}^p) + \sum_{t=1}^T \sum_{n=1}^{N_w} (c_{e-w} P_{t,n}^w) \quad (24)$$

2.2.3. Stability objective function

More and more renewable energy sources are connected to the power grid, because of their randomness and volatility, leading to thermal power units being in a long-term peak regulation state, and the resulting frequent startup and shutdown and ramp of unit are not conducive to the sustainable development of the power system. Meanwhile, with the revolution in the power system, the peak regulation task of TP will gradually shift. For catering to the future development trend of the power system, to use PS as risk management energy storage devices for multi-energy co-generation systems and to reduce the volatility of TP in the system. In this background, taking into account the volatility of the TP in the optimization operation model, set the objective function O_3 as follow

$$O_3 = \lambda \sum_{t=2}^T \sum_{n=1}^{N_f} |P_{f,t,n} - P_{f,t-1,n}| + \lambda \sum_{n=1}^{N_f} |P_{f,1,n} - P_{f,T,n}| \quad (25)$$

$$C_{c-out} = \begin{cases} p_{c-base}(Q_{c,r} - Q_{c,q}), & Q_{c,r} \leq Q_{c,q} + \Delta Q_c \\ p_{c-base}\Delta Q_c + (1+k)p_{c-base}(Q_{c,r} - Q_{c,q} - \Delta Q_c), & Q_{c,q} + \Delta Q_c \leq Q_{c,r} \leq Q_{c,q} + 2\Delta Q_c \\ (2+k)p_{c-base}\Delta Q_c + (1+2k)p_{c-base}(Q_{c,r} - Q_{c,q} - 2\Delta Q_c), & Q_{c,q} + 2\Delta Q_c \leq Q_{c,r} \leq Q_{c,q} + 3\Delta Q_c \\ (3+3k)p_{c-base}\Delta Q_c + (1+3k)p_{c-base}(Q_{c,r} - Q_{c,q} - 3\Delta Q_c), & Q_{c,q} + 3\Delta Q_c \leq Q_{c,r} \leq Q_{c,q} + 4\Delta Q_c \\ (4+6)p_{c-base}\Delta Q_c + (1+4k)p_{c-base}(Q_{c,r} - Q_{c,q} - 4\Delta Q_c), & Q_{c,q} + 4\Delta Q_c \leq Q_{c,r} \end{cases} \quad (23)$$

Box I.

2.3. Constraints

2.3.1. Load balance constraint

Electricity, as a special energy source that has been used only in recent centuries, has many characteristics, one of which is that power generation and power consumption are always done simultaneously, so the output of each power generation unit in the power system and the power consumption load should be balanced in real time, as shown in Eq. (26).

$$P_{load,t} = \sum_{n=1}^{N_f} P_{f,t,n} + \sum_{n=1}^{N_p} P_{t,n}^p + \sum_{n=1}^{N_w} P_{t,n}^w + \sum_{n=1}^{N_h} P_{t,n}^{hh} + \sum_{n=1}^{N_h} P_{t,n}^{hp} \quad (26)$$

2.3.2. Thermal power constraints

Combining the background of the new power system and the actual operating characteristics of TP unit, constraints are set as follows (Daneshvar et al., 2020; Bafrani et al., 2022)

$$U_{t,n}^f P_{min,n}^f \leq P_{f,t,n} \leq U_{t,n}^f P_{max,n}^f \quad (27)$$

$$P_{down,n}^f \leq P_{f,t,n} - P_{f,t-1,n} \leq P_{up,n}^f \quad (28)$$

$$P_{down,n}^f \leq P_{f,1,n} - P_{f,T,n} \leq P_{up,n}^f \quad (29)$$

$$T_{f-run} \geq T_{f-run}^{min} \quad (30)$$

$$T_{f-off} \geq T_{f-off}^{min} \quad (31)$$

Eq. (27) gives the upper and lower limits of the TP unit's output, Eq. (28) represents the thermal unit output ramp up/down limitation. Similar to the pumped storage units in the previous Section 2.1, Eq. (29) gives the start/end power limit. Since TP unit have high startup and shutdown cost, in order to avoid frequent startup and shutdown of TP unit, there is a constraint on the minimum startup and shutdown time of TP unit, as shown in Eq. (30) and (31).

2.3.3. Photovoltaic and wind power constraints

The output of wind power and photovoltaic unit are generally determined by wind resource, solar resource and unit (Wang et al., 2023a).

$$U_{t,n}^p P_{p-min,n} \leq P_{t,n}^p \leq U_{t,n}^p \min(P_{t,n}^{p-pre}, P_{max,n}^p) \quad (32)$$

$$U_{t,n}^w P_{w-min,n} \leq P_{t,n}^w \leq U_{t,n}^w \min(P_{t,n}^{w-pre}, P_{max,n}^w) \quad (33)$$

2.4. Research indicators

2.4.1. Thermal power volatility indicator

The power system revolution will make the proportion, role and status of TP in the power system change, accordingly, in the new energy environment, renewable energy gradually replace the TP output, pumped storage as a perfect reconciler to provide a guarantee for a large number of renewable energy consumption, in this context, TP should play a stable role in power generation. So the research proposes the TP volatility indicator as follow

$$I_v = \frac{2 \left[\sum_{t=2}^T \sum_{n=1}^{N_f} |(P_{f,t,n} - P_{f,t-1,n})| + |P_{f,1,n} - P_{f,T,n}| \right]}{TN_f (P_{up,n}^f - P_{down,n}^f)} \quad (34)$$

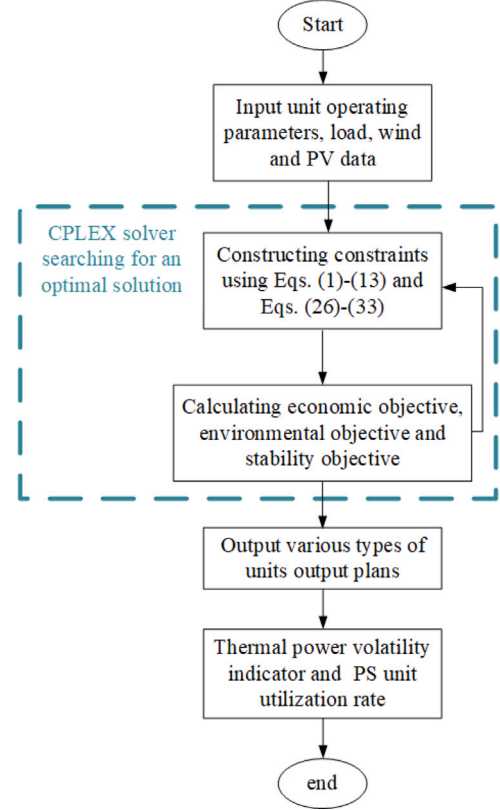


Fig. 2. System operation flow chart.

2.4.2. PS unit utilization rate

Reasonable capacity PS unit have a significant effect in the power system to improve renewable energy consumption and system flexibility. In order to study in depth the rationality of pumped storage capacity in the system, PS unit utilization rate is proposed, expressed as follow

$$I_{use}^h = \frac{\sum_{t=1}^T \sum_{n=1}^{N_h} (U_{t,n}^{hh} + U_{t,n}^{hp})}{TN_h} \quad (35)$$

3. Example studies

The system operation flowchart of this paper is shown in Fig. 2. Firstly, input the operating parameters of each unit, load data, wind power data and PV data. Secondly, the unit operation as well as the power generation system are constrained according to Eqs. (1)–(13) and 26–33, and the economic, environmental and stability objectives of the system are calculated according to Eqs. (14)–(25). Also, the CPLEX solver is called by MATLAB to find the optimal result according to the objective function. Finally, the operating results of each unit are output and calculated thermal power volatility indicator and pumped storage unit utilization rate.

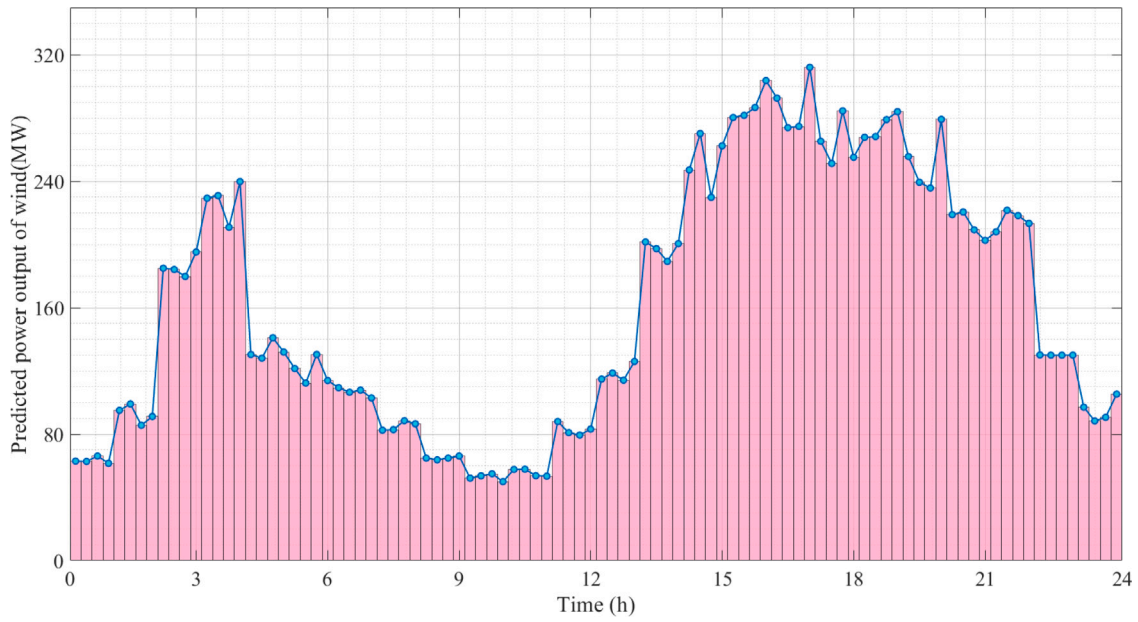


Fig. 3. Data diagram of wind power forecast output.

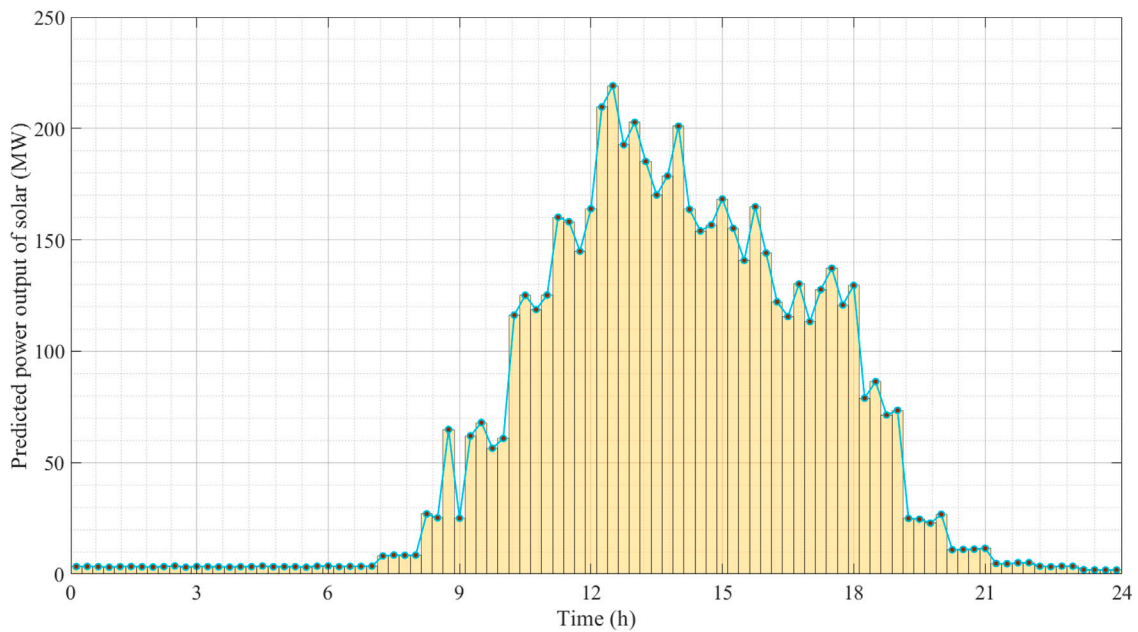


Fig. 4. Data diagram of photovoltaic power generation forecast output.

Table 1

Parameters for TP, PV and WP unit.

Characteristic parameters	TP	WP	PV
Rated power (MW)	420	320	220
Minimum stable generation (MW)	210	4.8	0

In this paper, the example of optimized operation of multi-energy co-generation system studied is based on a power system prototyped in a place in Gansu, China. The solar, wind power and electrical load data are chosen from the measured data of a typical day of summer in the area. The original data time scale is 1 h, which does not match the 15 min time scale studied in this paper, therefore, using random number to extend the data, the original data is used as the 3-th 15 min of each hour, and the 1-th, 2-th, and 4-th 15 min data are determined by

equation $p_{t,i} = p_{t,3} + 0.1yp_{t,3}$, where $p_{t,i}$ is the data for the t -th hour and the i -th 15 min, and y is a random number. The expanded data are used as example data, wind energy data, solar energy data, and electrical loads are shown in Figs. 3, 4 and 5. Fig. 6 is the pumping tariff for PS unit. The economic parameters of the research are shown in Appendix.

The example system consists of two TP units, two WP units, two PV units and four PS units, the specific parameters of which are shown in Table 1. The research sets the lower limit of the operating range of the PS units for power generation condition to be 40% of the rated power, and the ramp up/down to be 40% and -40% times of the rated power, respectively.

4. Results analysis

Based on engineering application level optimal operation model of WP-PV-TV-PS multi-energy co-generation system based on pumped

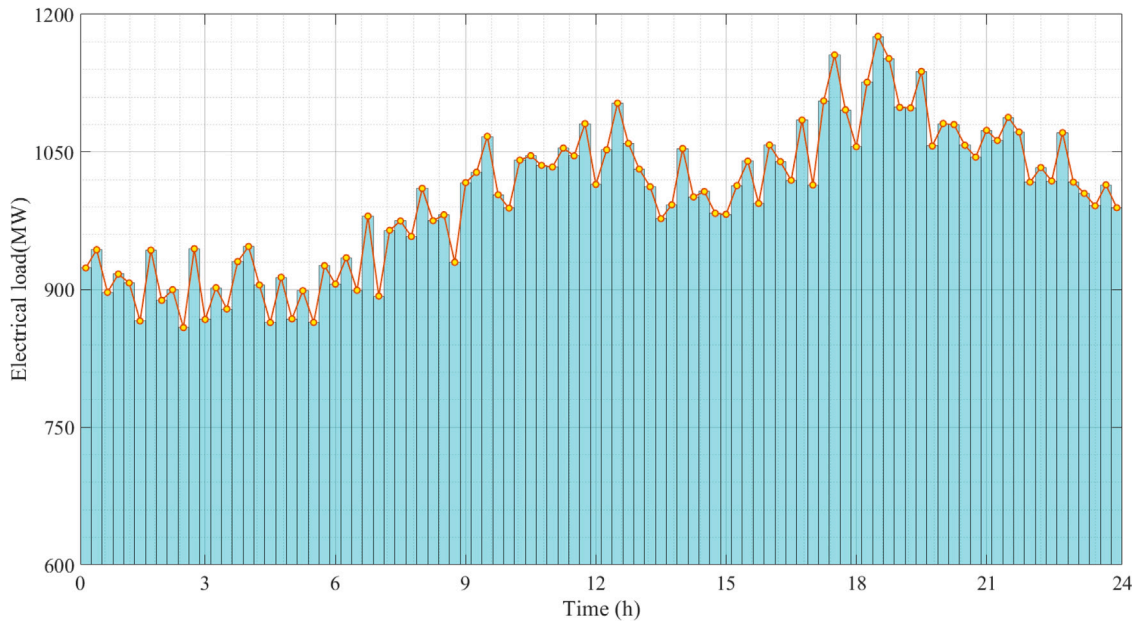


Fig. 5. Data diagram of electrical load.

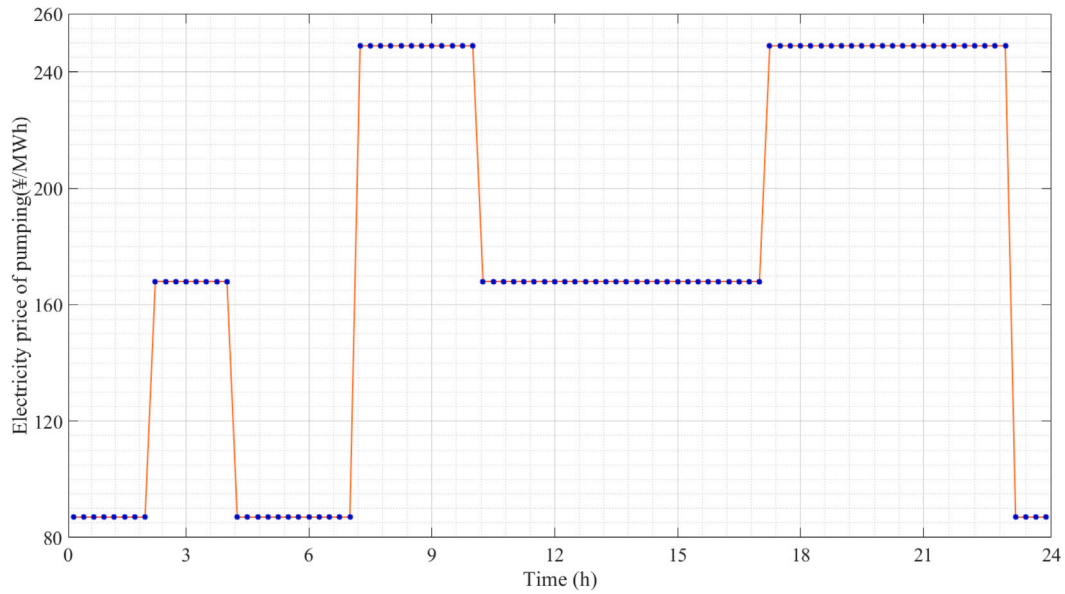


Fig. 6. Pumped storage unit pumping tariffs.

storage high fidelity modeling proposed in this paper, the example has been solved. Comparison and analysis of key data such as generation cost, environmental cost, unit combination, output distribution and pumped storage utilization under different design scenarios were carried out.

For analyzing the influence and difference between the operation model based on CPSM and the operation model based on HFPSM on the system operation, this paper calculates the pumped storage conventional model operation results and the high fidelity model operation results of the pumped storage unit with a rated capacity of 50 MW, respectively, and the operation results of the multi-energy co-generation system without pumped storage are used as a reference.

In addition, the study sets up a multi-energy co-generation system in which the single rated capacity of pumped storage units varies from 0 to 100 MW at intervals of 10 MW to analyze the effects of different pumped storage installed capacity on the indexes of the multi-energy co-generation system.

4.1. Impact of HFPSM on multi-energy co-generation system

For the purpose of understanding the optimization operation effect of HFPSM proposed in the paper relative to CPSM, Fig. 7 shows the daily output distributions of CPSM, HFPSM, and the optimization operation model without pumped-storage, respectively. As can be seen from the figure, renewable energy has significant volatility and has a relatively strong anti-peaking regulation effect for the power system. In Fig. 7(c), the system without pumped storage, due to the addition of renewable energy sources, TP unit need to frequently change operating condition and adjust the output power with large variation, which does not allow for continuous and stable production. Furthermore, the total power generation cost is 5962313.93 RMB, of which, the environmental cost is as high as 86987.14 RMB, while causing 13.3 wind curtailment. This indicates that the system without pumped storage caused large carbon emissions and renewable energy curtailment, which is

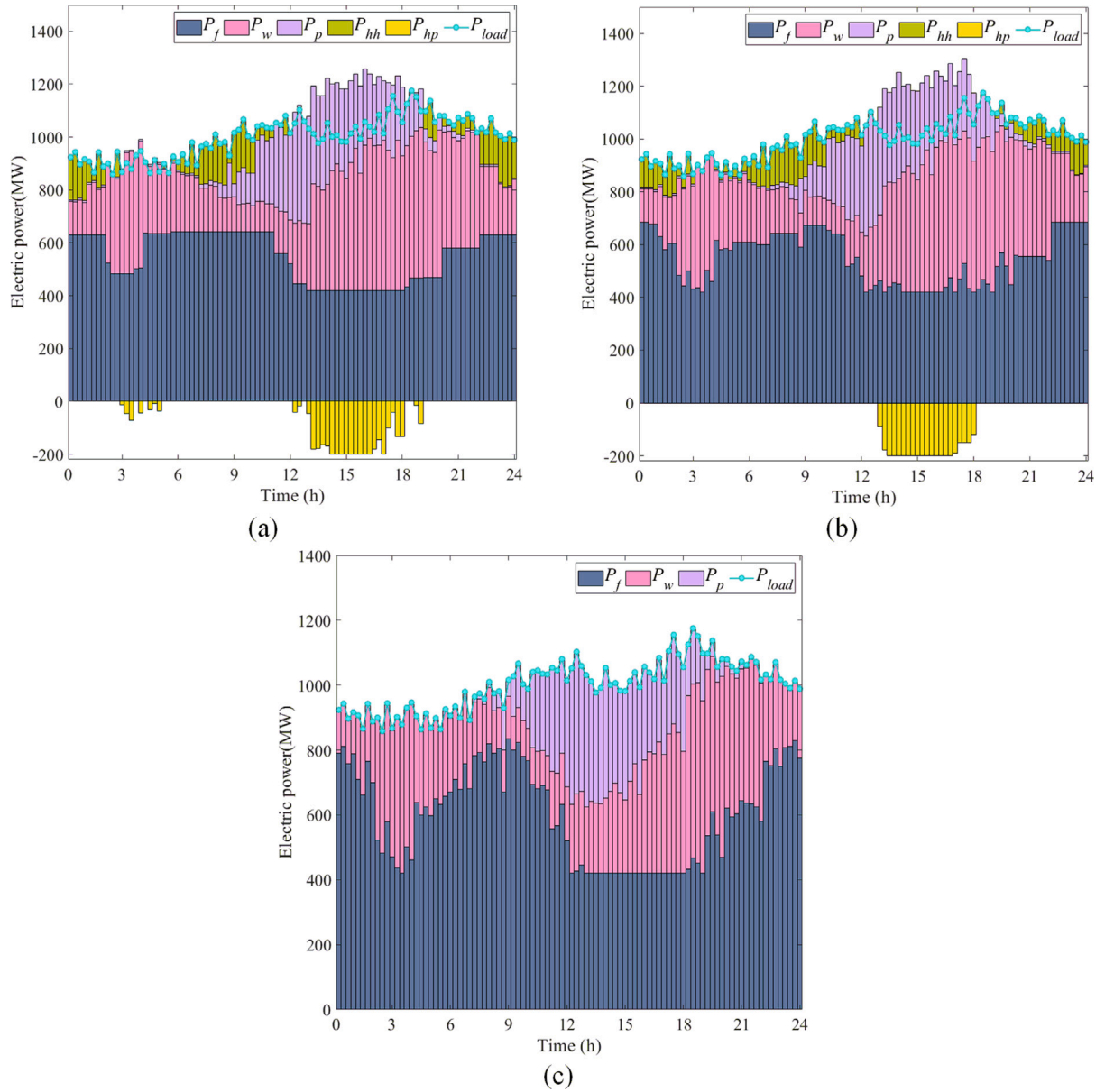


Fig. 7. Model optimized daily output distribution for each energy source, (a) CPSM, (b) HFPSM, (c) Without PS.

seriously inconsistent with the Sustainable Development Goals (SDGs) formulated by the United Nations. For the case of adding pumped storage, it can be seen from Fig. 7(a) and (b) that the situation is improved for both CPSM and HFPSM. The total generation costs for CPSM and HFPSM are 5236575.44 RMB and 5345940.98 RMB, which are 12.17% and 10.34% lower relative to without pumped storage. The environmental costs of CPSM and HFPSM are 12520.69 RMB and 14258.19 RMB, which are reduced by 85.61% and 83.61%, respectively, relative to without pumped storage. Also, the wind curtailment rate is significantly reduced, 1.8% and 2.1%. Meanwhile, frequent change of operating conditions and large change of TP output have been improved. However, these two are different, as can be seen from the figure, the HFPSM's thermal power volatility is slightly higher than CPSM's, this is due to the fact that, the consumption of renewable energy needs flexible capacity to smooth its volatility, in CPSM, the flexibility of the pumped storage unit is exaggerated. However, the HFPSM set up the non-operable operating condition regions, to improve the deficiency, ensure that the model's accuracy. In the actual operation of the system, the system does not have as much flexible capacity as described by CPSM, and the consumption of renewable energy requires

the cooperation of TP, as shown in Fig. 7(b). In addition, although PS acts as a risk management energy storage devices during operation, the safe and stable operation of the pumped storage unit cannot be ignored. In the pumped storage output results of CPS in Fig. 7(a), the power station operation time period is scattered, and there are many times of large variations in the adjustment of the output power. For the optimized operation results of the improved HFPSM in Fig. 7(b), the operation of the power station is relatively more centralized and there are no cases of large variations in the adjustment of the output power. Under the premise of ensuring the safe, stable and efficient operation of the system, HFPSM exhibits more consistent with the actual operation characteristics and better optimization results.

Fig. 8 shows the wind power consumption of CPSM, HFPSM, and the optimization operation model without pumped-storage. Since the model is set to prioritize PV consumption, there is little difference in PV consumption in the optimization results, so the wind power consumption results are used as a representative to show the renewable energy consumption. From the figure, it can be seen that the multi-energy co-generation system without pumped storage has the most wind curtailment rate with 13.3%, and after adding pumped storage,

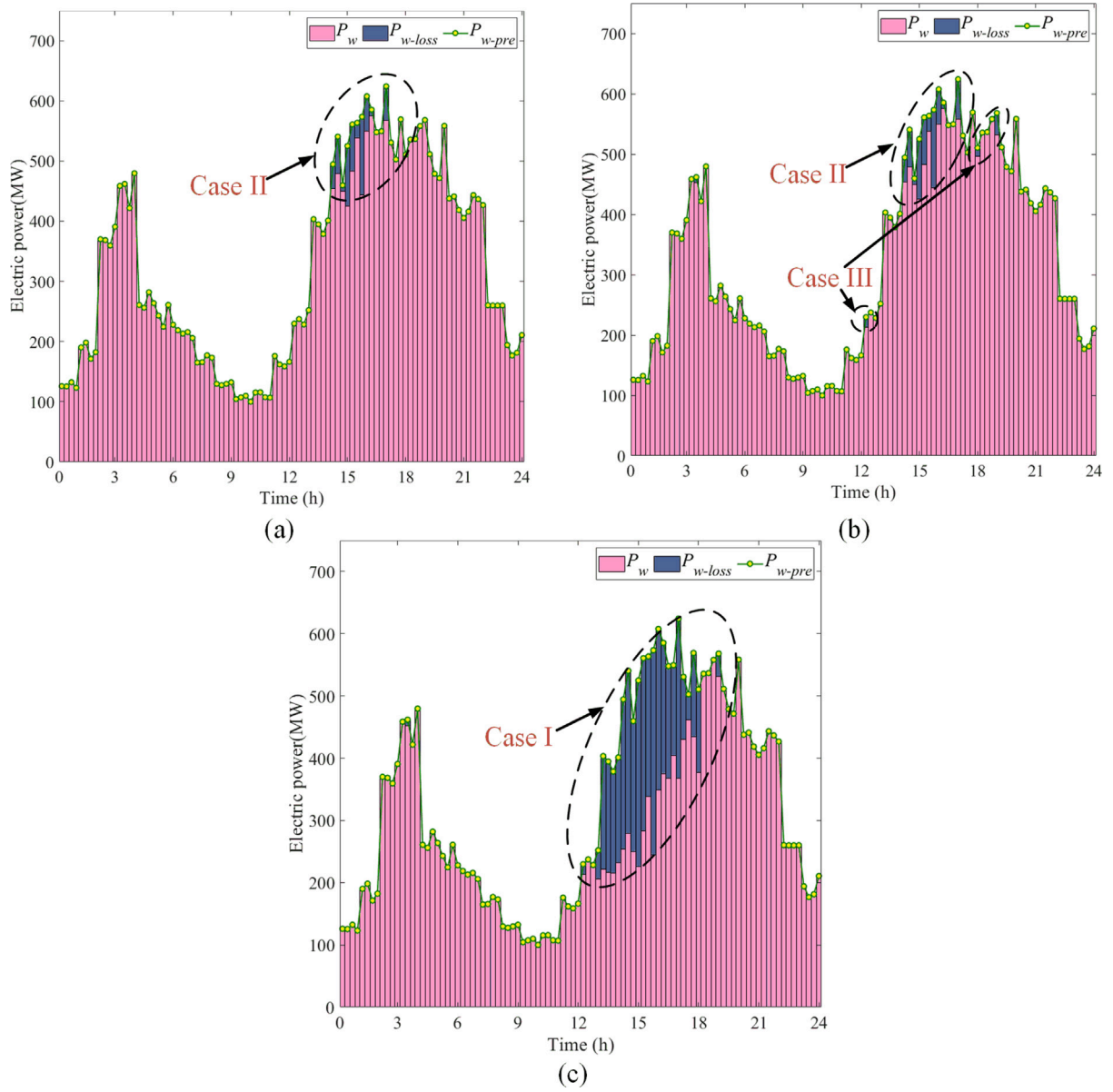


Fig. 8. Wind power consumption results, (a) CPSM, (b) HFPSM, (c) Without PS.

the system's wind curtailment rate is greatly reduced, and the least wind curtailment is in CPSM, followed by the HFPSM. Analyzing the three examples of wind curtailment mechanism, the wind curtailment was categorized into three cases. Case I appears in the results of the operation of a multi-energy co-generation system without PS, as shown in Fig. 8(c). This wind curtailment is consistent with the situation in many areas where a large number of new renewable energy stations are now being built, where the system has little or no energy storage, which leads to insufficient flexible regulation capacity in the power generation system, causing large amount of wind curtailment. Case II occurs in both CPSM and HFPSM operation results, as in Fig. 8(a) and (b), in this scenario, the power generation system is set up with energy storage, but due to the extreme abundance of renewable energy sources, it exceeds the scope of the system's ability to consume them, and this case can be solved by increasing the system's energy storage capacity. Case III appears in the operation results of HFPSM, this wind curtailment phenomenon occurs because the HFPSM, compared with the CPSM, corrects the inadequacy of CPSM for the exaggerated description of the flexibility of pumped storage, so that the optimization results are more in line with the actual operation of the unit and ensure the accuracy

of the model. Therefore, compared with the results of CPSM, the wind curtailment phenomenon occurs in the 49-th, 72-nd, and 76-th time period, which can be solved by adding chemical energy storage or other flexible energy storage method.

Fig. 9 shows the daily output results of four pumped storage units for CPSM and HFPSM, where the red area indicates that the unit is generating, the blue area indicates pumping, and the gray area indicates shutdown. From Fig. 9(a), it can be seen that the pumped storage units operation results of the CPSM are more chaotic, with the frequent occurrence of startup and shutdown, large variability of output power adjustment, and in many places, continuous pumping and power generation for only one period of time. Many places inconsistent with the actual operation rules of the units. Such as the fourth unit's generating condition in the time periods of 3-th, 29-th, and 33-th, pumping condition in the time periods of 15-th, 17-th and 48-th, etc., which are unacceptable for the pumped-storage operation results. In the operation results of pumped storage units in HFPSM, the undesirable operation that occurs in CPSM is improved, as shown in Fig. 9(b). For example, the number of switch-on/off times of pumped storage units in one operation cycle of CPSM is as high as 93, and the number of

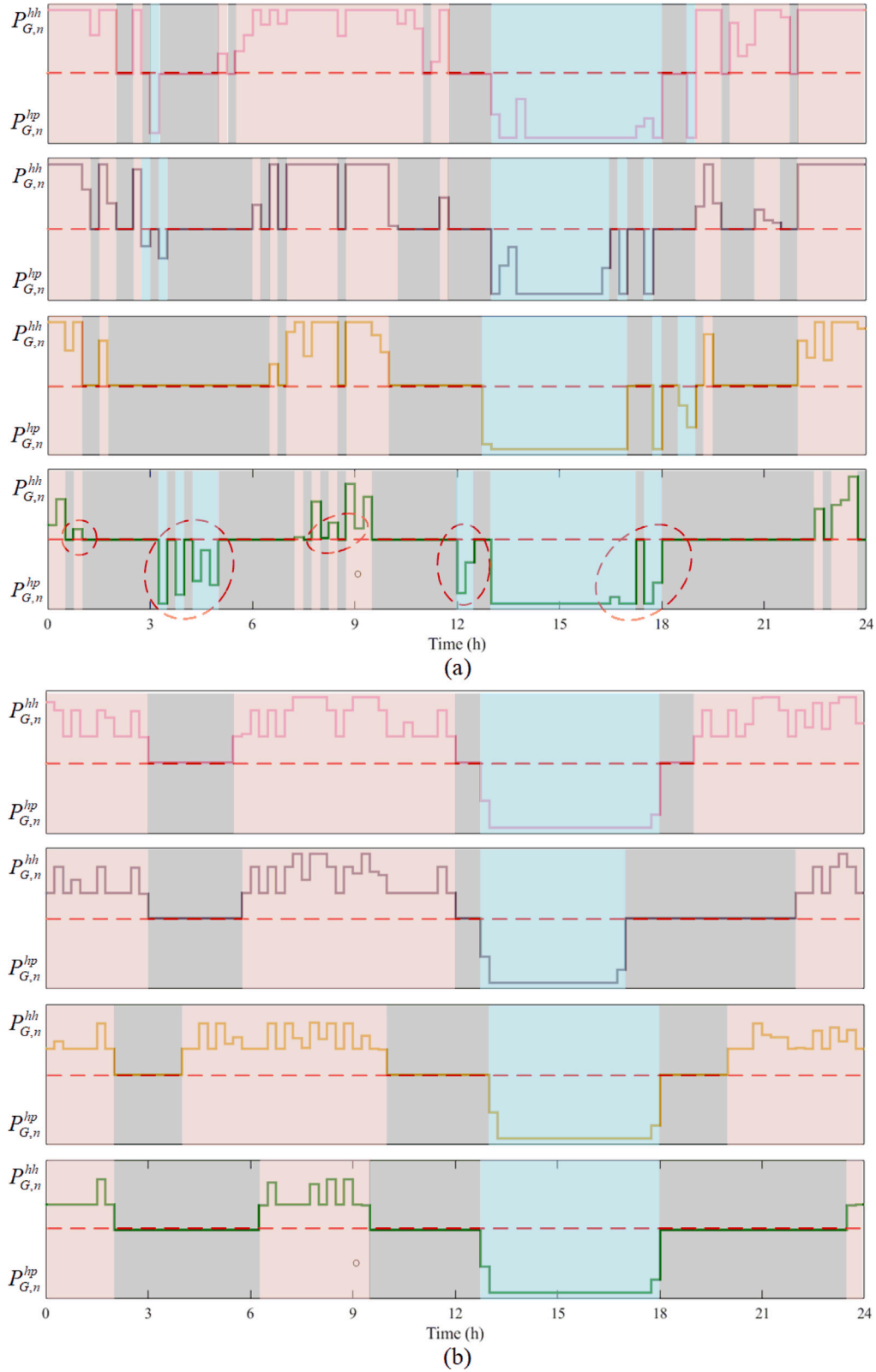


Fig. 9. Daily output results for four pumped storage units, (a)CPSM, (b) HFPSM.

switch-on/off times of the HFPSM is only 24, the number of switch-on/off times of the units in HFPSM is only 25.8% of that of CPSM. In addition, compared to CPSM in which the unit operates continuously for only one time period (15 min), the unit operation status of HFPSM is greatly improved. As can be seen in Fig. 9(b), the shortest continuous

operation time of the pumped storage unit is 135 min. Moreover, HFPSM takes into account the power loss of the unit in switching on and off, which is particularly evident in the pumping condition in Fig. 9(b), the operation results are more accurate. The operation results of pumped storage units in HFPSM are more in line with the actual

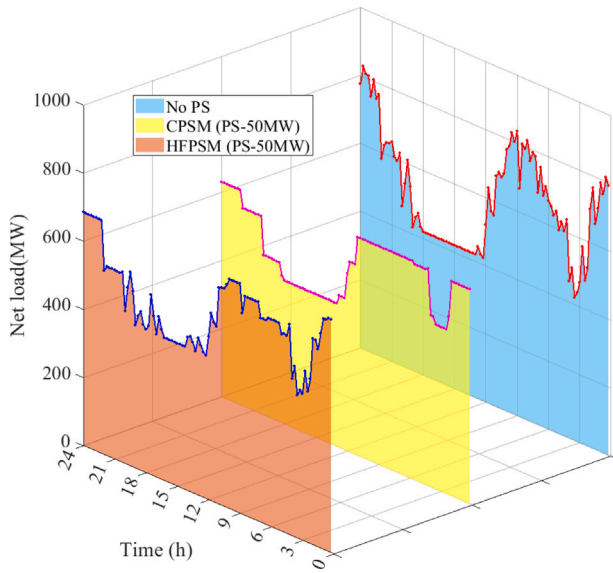


Fig. 10. Net load curves without pumped storage, CPSM and HFPSM optimized operation results.

operating characteristics of the unit under the premise of taking up the task of risk management energy storage.

Fig. 10 shows a comparison of the net load curves of the optimization results for the three cases. In the case without PS, the maximum peak-to-valley difference in the net load curve of the grid is 414 MW, with the addition of PS, the maximum peak-to-valley difference is reduced to 220 MW for CPSM and 265 MW for HFPSM. It can be seen that the addition of pumped storage to the multi-energy co-generation system can significantly reduce the peak-to-valley difference of the load, thus realizing “peak-cutting and valley-filling”. The net load curve of HFPSM is slightly cluttered relative to CPSM, and the maximum peak-to-valley difference is slightly increased, which corresponds to the previous discussion that HFPSM corrects for the inadequacy of CPSM’s exaggerated description of pumped-storage flexibility.

4.2. Impact of different PS installed capacities on multi-energy co-generation system

Energy storage capacity in multi-energy co-generation system is a key issue in power supply planning, different storage capacity will obviously have different impacts on the economy, stability, low carbon and renewable energy consumption of the power generation system. In this paper, the optimal operation of multi-energy co-generation systems with different pumped storage installed capacities is investigated, and the results are shown in Table 2. From the table, it can be seen that as the installed capacity of pumped storage increases, the cost of electricity generation and the wind curtailment rate generally tends to decrease, but it also fluctuates. The optimal total operating cost O , which is 5962313.93 RMB, is obtained at an installed pumped storage capacity of 100 MW, which is 11.55% lower than that of the generation system without pumped storage. The minimum generation cost O_1 is 5148707.09 RMB, obtained at an installed pumped storage capacity of 70 MW, with a reduction of 9.30% with respect to the generation system without pumped storage. The maximum environmental cost O_2 is reduced by 99.44% with a value of 489.84 RMB, corresponding to a pumped storage capacity of 90 MW, while the minimum wind curtailment rate is achieved at 0.27%. Thermal power volatility is minimized when the pumped storage capacity is 100 MW, at which time the stability cost O_3 and volatility rate I_v are 120711.50 RMB and 9.98%, respectively. The minimum utilization rate I_{use}^h of pumped storage is 40.89%, which corresponds to an installed capacity of 100 MW. The

reason for this phenomenon is that HFPSM improves the operating range of the unit and corrects the inadequacy of the previous CPSM that exaggerates the flexibility of the pumped storage unit. The result is that the flexibility of the pumped storage unit varies with the installed capacity, and the costs and indicators are constantly changing.

For detailed research on the phenomenon of volatility of the results appearing above with the change of installed capacity of pumped storage, we plot the changes in pumped storage units utilization rate, TP volatility rate, carbon emission costs and environmental benefits with the changes in pumped storage installed capacity, as shown in Fig. 11. Since the trends of total operating costs, power generation costs and carbon emission costs are similar, therefore take the carbon emission costs as a representation to analyze. As can be seen from Fig. 11(a) and (b), the utilization rate and the installed capacity of pumped storage units are not linearly related, but the sharpness and slowness of the declining trend in the utilization rate of pumped storage units are closely related to the other three indicators. When the installed capacity of pumped storage units changes from 10 MW to 20 MW, the pumped storage utilization rate decreases sharply. This is due to the 9–16th time period, wind power resources are abundant, in order to consume more renewable energy, when the pumped storage installed capacity is 10 MW, a pumped storage unit is pumped in this time period, as shown in Fig. 11(c). But at this time, the thermal power units in the 14th time period are already close to the minimum stabilized power, when the installed capacity increases, there is no excess power for the pumped storage units to consume, in combination with the unit’s minimum continuous operation time constraint. Therefore, when the installed capacity of the pumped storage unit increases to 20 MW, the pumped storage unit no longer works at this time, and the utilization rate of pumped storage drops sharply. During the change in pumped storage installed capacity from 20 MW to 50 MW, the unit utilization rate is relatively stable, and during this period, TP volatility rates, carbon emission costs drop sharply and environmental benefits rise sharply, corresponding to the sharp drop in total operating costs, generation costs and wind curtailment rate in Table 2. When the installed capacity of pumped storage units changes from 50 MW to 90 MW, the utilization rate decreases sharply, the carbon emission costs, TP volatility rates and environmental benefits in this process level off with some minor volatility. When the installed capacity of pumped storage units changes from 60 MW to 80 MW, there is an increase in TP volatility rates, and reaching a localized maximum at a pumped storage capacity of 80 MW. Fig. 11(d) shows the distribution of daily output for an installed capacity of 80 MW of pumped storage. The increased volatility is due to the fact that the grid load is not sufficient to consume the renewable energy in some small time periods within the 54–70th time period, at this time, pumping storage units are required to pump water. However, the HFPSM proposes non-operable operating condition regions, which limits the pumps to pumping only at their rated power. The unavailability to be consumed and the extra energy is less than the rated pumping power, the missing energy for pumping work of pumped storage units is replenished by the TP units, thus resulting in the increase of volatility rate in TP units. When the installed capacity of pumped storage is increased again, the TP supplemental energy makes pumped storage units pumping water to consume renewable energy bring benefits less than the increased cost, there will be a drop in unit utilization rate.

5. Conclusion

This paper improves the insufficiency of the CPSM in exaggerating the flexibility of pumped storage units by setting non-operable operating condition regions. Also, condition switching loss is taken into account in the model. On the basis of which a high fidelity model for pumped storage is established. In addition, based on three considerations of economy, environmental protection and stability, an engineering application level optimal operation model of multi-energy

Table 2

Results of optimized operation of multi-energy co-generation systems with different pumped storage installed capacities.

Installed capacity of PS units(MW)	O (RMB)	O_1 (RMB)	O_2 (RMB)	O_3 (RMB)	I_{w-c} (%)	I_{use}^h (%)	I_v (%)
0	5962313.93	5676370.39	86987.14	198950.91	13.30	\	16.45
10	5797726.64	5562413.75	68981.37	166331.51	10.55	80.99	13.75
20	5655381.24	5457019.41	53567.85	144793.99	8.12	69.79	11.97
30	5526741.75	5357038.45	38376.83	131326.47	5.77	68.23	10.86
40	5417392.40	5269447.38	23993.71	123951.31	3.60	68.23	10.25
50	5345940.98	5209704.75	14258.12	121978.12	2.13	65.89	10.08
60	5296390.82	5168025.45	7237.94	121127.43	1.11	60.16	10.01
70	5276773.72	5148707.09	4513.75	123552.88	0.70	53.39	10.21
80	5282311.78	5152577.45	2445.99	127288.34	0.49	49.74	10.52
90	5274542.37	5149279.36	489.84	124773.17	0.27	48.44	10.32
100	5273830.83	5150263.23	2856.10	120711.50	0.56	40.89	9.98

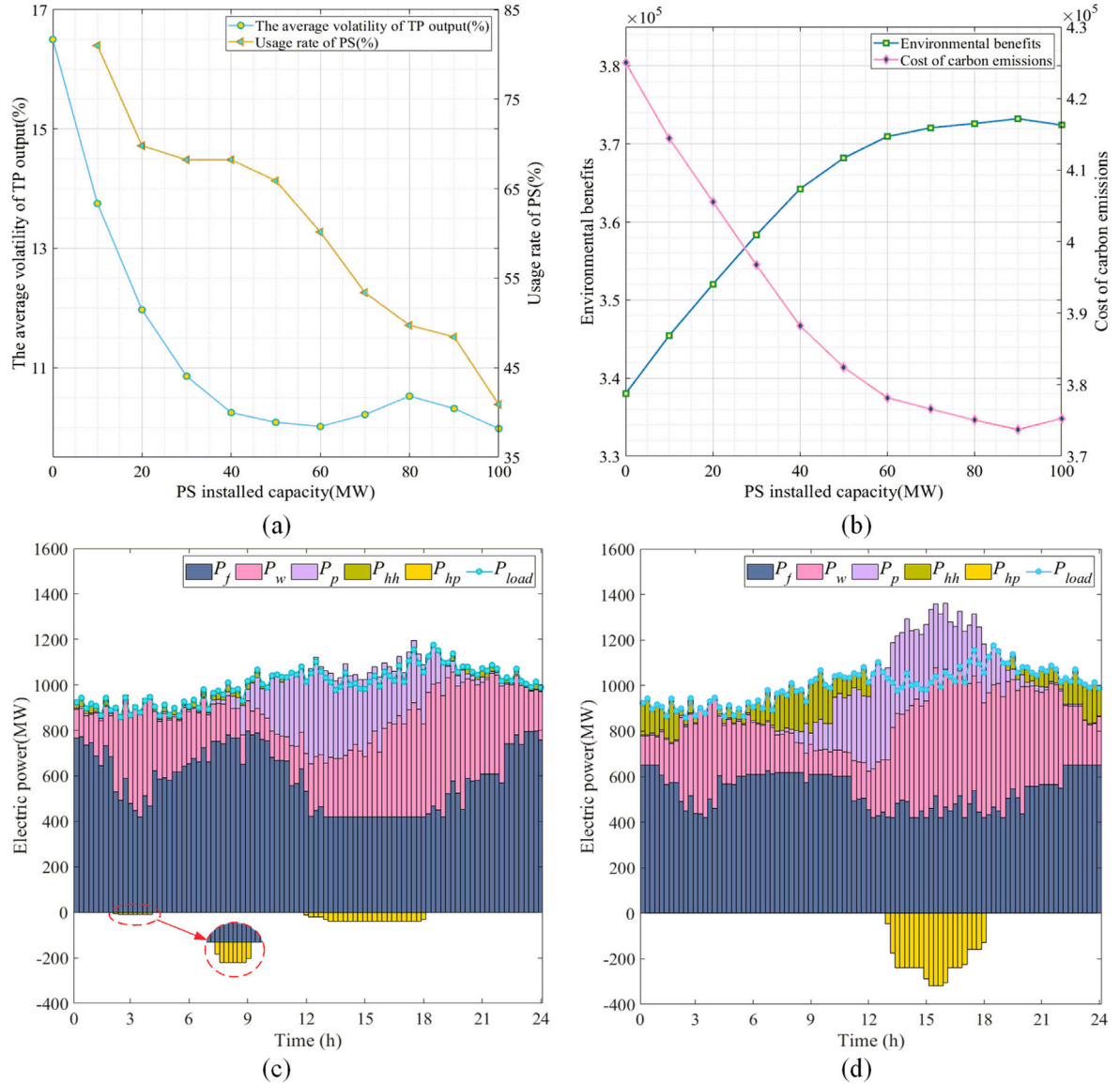


Fig. 11. (a) Change in average TP volatility rates and PS unit utilization rates with changes in installed capacity of PS units, (b) Change in environmental benefits and carbon costs with changes in installed capacity of PS units, (c) Daily output distribution for an installed capacity of pumped storage of 10 MW, (d) Daily output distribution for an installed capacity of pumped storage of 80 MW.

co-generation system based on HFPSM is established. Finally, the model is simulated and investigated by examples.

The results show that with the addition of four 50 MW pumped storage units to the system can reduce the total operating cost by 10.34% and the environmental cost by 83.61%. On the other hand, the CPSM exaggeratedly describes the flexibility of PS in the system.

The optimized operation results are inconsistent with the actual operation characteristics of pumped storage in many places, and there are frequent changes in operating conditions and large variations in the adjustment of output power. The proposed HFPSM corrects these deficiencies by improving the model so that the pumped storage unit is regulated in a stable and feasible regional range. Meanwhile, the

number of unit startup and shutdown is greatly reduced, and the number of HFPSM is only 25.8% of the CPSM. The HFPSM shows more consistent with the actual operation characteristics and better optimization effect under the premise of ensuring the safe, stable and efficient operation of the system.

By changing the installed capacity of pumped storage connected to the multi-energy co-generation system, the optimal operation of multi-energy co-generation systems with different installed capacities of pumped storage was investigated. It is found that the indicators of TP volatility rates, carbon emission costs and environmental benefits did not always continue to decrease with the increase in installed capacity of PS units, but fluctuated at a certain level of increase in pumped storage capacity. This is due to the fact that the HFPSM corrects the inadequacy of the CPSM's exaggerated description of pumped storage flexibility. The flexibility of the pumped storage units no longer increases with the increase of installed capacity, and the optimization results are more in line with the actual operation. So, the proposed HFPSM provides a reference basis for energy storage capacity planning in multi-energy co-generation systems.

In this paper, the economic objectives include various costs of system operation, but do not include the construction costs of the power station. Therefore, in the study of changing the installed capacity of pumped storage, there is no obvious optimal pumped storage capacity in terms of economy. The pumped storage capacity that is relatively friendly to the system can only be judged by combining various indicators to be 50–60 MW. In future studies, the construction cost of the power station should be fully considered and the model should be described more comprehensively.

CRediT authorship contribution statement

Ganggang Liang: Formal analysis, Methodology, Visualization, Writing – original draft. **Hao Zhang:** Conceptualization, Supervision, Writing – review & editing. **Chenxi Li:** Investigation, Methodology, Resources. **Xin Wu:** Visualization. **Pengcheng Guo:** Supervision, Writing – review & editing. **Haipeng Nan:** Resources, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix

The economic parameters studied are shown in Table A.1.

Table A.1

Economic parameters.

Parameters	Value	Parameters	Value
c_{f2}	0.052(RMB/MW ²)	c_{f1}	216(RMB/MW)
c_f	2483(RMB/h)	c_p	55(RMB/MW)
c_w	50(RMB/MW)	$c_h h$	70(RMB/MW)
c_{m-f}	70(RMB/h)	c_{m-p}	34(RMB/h)
c_{m-w}	29(RMB/h)	c_{m-h}	250(RMB/h)
c_{p-pen}	500(RMB/MW)	c_{w-pen}	550(RMB/MW)
p_{c-base}	140(RMB/t)	c_{e-p}	35(RMB/MW)
c_{e-w}	35(RMB/MW)	λ	50(RMB/MW)

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